Davide Modolo

AI & MACHINE LEARNING ENGINEER | MSC ARTIFICIAL INTELLIGENCE

AI & ML Engineer aiming to own models end to end: train, fine-tune, deploy, and serve at scale.

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Work Experience

OpenCity Labs - *Startup*

Remote

AI & MACHINE LEARNING ENGINEER - R&D

August 2025 - Current

- Developed, fine-tuned, and benchmarked a spaCy multilingual NER model for PII detection, deployed in-process in the production service and used on Italian citizen messages; error analysis over real queries improved precision on names, locations, and organizations.
- Engineered the chatbot pipeline: 50k+ public-administration pages ingested daily, filtered through the NER model and regex rules into an allowed-list; user messages pass the same layer, which anonymizes entities outside it while preserving meaning.
- Reduced a cross-lingual failure from 35% to 1% of non-Italian queries answered in Italian by adding answer-language mismatch detection with LLM translation fallback.
- Co-designed an open-source agentic AI platform for municipal officers: an orchestrator coordinates specialized subagents via MCP tool-calling, with human-in-the-loop approval on proposed actions.
- Centralized AI automation flows in Windmill, an open-source orchestration framework, exposing some as MCP tools to the agent.

`LLMs, RAG, NER, MCP, AWS Bedrock, Windmill, FastAPI, Docker, GitLab CI`

Eurecat Technology Center

Barcelona, Spain

MACHINE LEARNING RESEARCH INTERN

April 2024 - June 2024

- Implemented KnowNo, an uncertainty alignment framework for LLM planners (from Google's *Robots That Ask For Help* paper): top-20 token log-probability analysis across GPT-4o and locally quantized Llama 3.1 (llama.cpp), delivered as a reusable Python package.
- Developed an uncertainty-driven action-selection loop for an elderly-care robotic assistant: when multiple actions pass the calibrated threshold, the planner re-prompts with only the most probable options or asks the user, trading autonomy for safety; outcomes stored via RAG (*gte-large*).
- Extended the framework beyond text: tested LLaVA on photos of real-life scenes to bring visual input into uncertainty estimation, validating the approach across ∼10 simulated environments.

`PyTorch, GPT-4o, Llama 3.1 (llama.cpp), RAG, Uncertainty Quantification, LLaVA`

University Projects

COVID-19 Lung Ultrasound Images classification - Medical Imaging Diagnostic

[slides | report | code]

- Designed a multi-stage pipeline for COVID-19 LUS severity scoring (0-3): a fine-tuned ResNet18 frame classifier, a binary SVC uncertainty detector on softmax outputs, and a t-SNE-based similarity module retrieving nearest neighbours for low-confidence predictions.
- Trained and tested on 47k anonymized frames from 11 patients with patient-level splits; augmentation (affine, blur, flips) combated overfitting.

`ResNet18, scikit-learn, Medical Imaging, Computer Vision`

Joint Intent Detection and Slot Filling - Natural Language Understanding

[report | code]

- Compared 4 architectures for joint intent detection and slot filling, from fine-tuned pre-trained transformers (BERT, ERNIE) — which struggled with vocabulary mismatch on custom slot tokens — to recurrent models built from scratch (BiLSTM, Encoder-Decoder).
- Custom encoder-decoder achieved the best results, 94.95% slot F1 and 96.42% intent accuracy on ATIS, with +2%/+13% gains over baselines.

`NLP, BERT, ERNIE, BiLSTM, Encoder-Decoder`

Domain Adaptation / Transfer Learning - Deep Learning - *Team Project*

[code]

- Built an unsupervised domain adaptation model based on the HoMM paper: an ImageNet-pretrained ResNet34 with a custom adaptation layer using direct 3rd-order and grouped 4th-order statistics to bridge the synthetic-to-real gap.
- Achieved +11.53% improvement over the non-adapted baseline on the Adaptiope dataset (20 classes, product_images → real_life).

`ResNet34, HoMM, Transfer Learning`

Education

University of Trento

Trento, Italy

Master's Degree in Artificial Intelligence Systems

September 2021 - Graduated March 2025

- **Thesis**: Exploring the Use of LLMs for Agent Planning: Strengths and Weaknesses. *[report | code]*
- **Relevant Courses**: Machine Learning, Deep Learning, Natural Language Understanding, Automated Planning, Law & Ethics in AI, Fundamentals of AI (Reasoning and Planning), Signal Processing, AI for Finance, High Performance Computing.

University of Trento

Trento, Italy

Bachelor's Degree in Computer Science

September 2017 - Graduated June 2021

- **Core Topics**: Algorithms & Data Structures, Software Engineering, Database Systems, Operating Systems, Computer Architectures, Distributed Networks.

Skills

Languages	`Python, JavaScript, C/C++, SQL, Bash, LaTeX`
ML & Data	`PyTorch, Hugging Face, Transformers, Deep Learning, Fine-tuning, LLMs, RAG, NLP, Computer Vision`
DevOps & Cloud	`Git, Docker, GitLab CI, FastAPI, Deployment & Serving, AWS (EC2, Bedrock), Grafana, Prometheus`
Spoken Languages	**English** - Professional, **Italian** - Native
OSS Contributions	`Windmill`[1], `cc_scrapycat`[1234], `ccat-dietician`[1]